PREPARED BY VU SQUAD 0313-7248280
LMS AND PROJECTS HANDLER EXPERT



MATH-622 FINAL TERM QUIZES BY VU SQUAD

LMS HANDLING

ASSIGNMENT

QUIZ

GDB

LECTURE WATCH

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PROJECTS

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Question:

The restoring force of a harmonic oscillator is proportional to .....

Options:

A. acceleration

B. displacement

C. time

D. velocity

Question:

According to Newton's second law of motion the net force applied on a body is equal to the rate of change of momentum of body.

Options:

A. true

B. false

Question:

In case of conservative force field, the total energy is.....

Options:

A. a constant

B. any function of time

Question:

If a force acts on a particle of constant mass m and \vec{v}_{1} , \vec{v}_{2} are velocities at times t_{1} and t_{2} , respectively. Then the work done by the force is obtained as

Options:

A. The change in kinetic energies, i.e., \frac{1}{2}m\vec{v}_{2}^{2}-\frac{1}{2}m\vec{v}_{1}^{2}

B. The sum of kinetic energies, i.e., \frac{1}{2}m\vec{v}_{2}^{2}+\frac{1}{2}m\vec{v}_{1}^{2}

Question:

Let 56kgms -1 be the linear momentum of a system of particles and 4ms -1 be the velocity of the center of mass then the total mass of the system would be _____kg.

Options:

A. 120

B. 68

C. 14

D. 46

Question:

The force field \vec{F}=3x\hat{i}-2y\hat{\hat{j}} is

Options:

A. conservative

B. non conservative

Question:

Find the time period T of the vibrating mass with m=5 and f=10

Options:

A. T=\frac{1}{10}

B. T=\frac{1}{2}

Question:

If a particle of mass 2 units has position vector \vec{r}=t^{3}\hat{i}+8t\hat{j}-9t\hat{k} then find the force acting on the particle.

Options:

A. 12t

B. 12t\hat{i}

Question:

The product of force and displacement is called

Options:

A. energy

B. work done

Question:

Let 16kgm 2 be the momentum of inertia of a system of particles about AB, and 4kg be the total mass of the system then the radius of gyration of the system AB would be _____m.

Options:

A. 2

B. 8

C. 4

D. 1

Question:

If no external force is acting on the system of two colliding bodies, then by the principle of conservation of linear momentum which of the following statement is correct?

Options:

A. Total momentum before the collision is greater than the total momentum after the collision.

B. Total momentum before the collision is less than the total momentum after the collision.

C. Total momentum before the collision is equal to the total momentum after the collision.

D. Total momentum before the collision is reciprocal of the total momentum after the collision.

Question:

Let 0.01kgm 2 be the moment of inertia (about an axis passing through the fixed point) of a rigid body rotating with the angular velocity 20 rad/s. Its kinetic energy will be ____ J.

Options:

A. 1

B. 5

C. 4

D. 2

Question:

A solid metal sphere having the moment of inertia of 0.03kgm 2 is rotating with an angular speed of 13.8 rad/s. Its angular momentum would be _____kgm 2 s -1

Options:

A. 0.440

B. 0.354

C. 0.414

D. 0.204

Question:

Let 5kg be the mass and 0.1m be the radius of a solid sphere. Its moment of inertia about the diameter will be ____kgm 2

Options:

A. 0.10

B. 0.02

C. 0.01

D. 0.05

Question:

If a system is isolated, then the total external torque (moment of force) remains constant in _____.

Options:

A. both magnitude and direction

B. magnitude

C. direction

D. number of particles of the system

Question:

Let 64kgms -2 be the rate of linear momentum of a particle moving with the acceleration 16ms -2 then mass of the particle would be _____kg.

Options:

A. 28

B. 14

C. 4

D. 8

Question:

Let 1.2kg be the mass and 3m be the length of a thin solid rod. Its moment of inertia about the axis perpendicular to the rod through the center of mass will be ____kgm 2

Options:

A. 3.6

B. 1.8

C. 5.4

D. 2.7

Question:

Let P be any point on the ellipsoid of inertia and 0.25kgm 2 be the moment of inertia about any line OP of the rigid body. The length of line OP will be _____m.

Options:

A. 2

B. 5

C. 4

D. 1

Question:

A solid metal sphere having the moment of inertia of 0.44kgm 2 is rotating with an angular speed of 2 rad/s.Its rotational kinetic energy would be _____ J.

Options:

A. 0.88

B. 0.66

C. 0.48

D. 0.76

Question:

What happens to angular velocity when the moment of inertia of an isolated system is doubled to keep the angular momentum constant?

Options:

A. It is halved.

B. It becomes zero.

C. Remains same.

D. It doubles.